Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP210LT001-V1

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Revision Record

Rev No.	Date	Description
V.0	2026-04-02	First Release

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Revision Record

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1.General Specifications

No.	Item	Specification	Unit	Remark
1	LCD Size	2.1	inch	-
2	Panel Type	IPS_incell	-	-
3	Resolution	480x RGB x 480	Pixel	-
4	Display Mode	Normally Black		-
5	Number of Colors	16.7M	-	-
6	Viewing Direction	All	-	Note1
7	NTSC	69%	-	Typ.
8	Contrast Ratio	1000	-	
9	Luminance	250	cd/m2	Typ.
10	Module Size	66.7(H)x66.7(V)x4.04(D)	mm	Note1
11	Panel Active Area	53.28x53.28	mm	Note1
12	Pixel Pitch	0.425(H) x 1.275 (V)	mm	-
13	Pixel Arrangement	RGB-stripe		-
14	Weight	TBD	g	-
15	Driver IC	ST77922	-	-
16	Driver IC RAM Size		bit	-
17	Light Source	4white LEDs	-	-
18	Interface	MIPI+I2C	-	-
19	Operating Temperature	-20~+70	℃	-
20	Storage Temperature	-30~+80	℃	-

Note 1: Please refer to the mechanical drawing；

注 1：请参照模组图；

2.Pin Assignments

Pin No. Pin 序号	Symbol 符号	Function 功能描述
1	NC	No connection
2	NC	No connection
3	GND	Ground
4	GND	Ground
5	D0N	MIPI DSI data lane0-
6	D0P	MIPI DSI data lane0+
7	GND	Ground
8	GND	Ground
9	CLKN	MIPI DSI data lane-
10	CLKP	MIPI DSI data lane+
11	GND	Ground
12	TP_SDA	I2C SDA,TYP. 1.8V
13	TP_SCL	I2C CLK,TYP. 1.8V
14	RESET	LCM reset signal
15	TP_RST	The external reset input for TP
16	TP_INT	Interrupt signal for TP
17	IOVDD18	Power Supply 1.8V
18	VDD-3V3	Analog Power Supply for LCM 系统输入电压
19	GND	Ground
20	TP-VDD/NC	
21	GND	Ground
22	LEDK	Backlight cathode
23	LEDA	Backlight anode
24	GND	Ground

3. Electrical Specification

3.1Absolute Maximum Ratings

Item 项目	Symbol	Value	Unit	Remark
Analog Power Supply Voltage 模拟供电电压	VCI	-0.3~+3.3	V	-
Digital Power Supply Voltage 数字电源电压	VDD	-0.3~+3.3	V	-
I/O Power Supply Voltage I/O端口供电电压	IOVCC	-0.3~3.3	V	-

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3. Electrical Specification

3.2 Typical Operation Conditions

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Supply Voltage 模拟供电电压	VCI	2.7	2.8	3.3	V
Digital Supply Voltage 数字电源电压	VDD	2.7	2.8	3.3	V
I/O Supply Voltage 接口电压	IOVCC	1.65	1.8/2.8	3.3	V
Input High Voltage 输入高电平	V_{IH}	0.8*IOVCC	-	IOVCC	V
Input Low Voltage 输入低电平	V_{IL}	0	-	0.2*IOVCC	V
Output High Voltage 输出高电平	V_{OH}	0.8*IOVCC	-	-	V
Output Low Voltage 输出低电平	V_{OL}	-	-	0.2*IOVCC	V

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3.3 Backlight Circuit Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B	-	80	-	mA
LED Voltage 背光电压	Vf		3.2		V
Power Consumption 功耗	P_{BL}	-		-	mW

3.4 Power Consumption

Item	Symbol	Typ.	Max.	Unit
Full Mode 正常模式	VCI	-	-	mA
测试条件：VCI=2.8V，IOVCC=1.8V； Interface 驱动类型：行翻转或者列翻转； TN Type=>All Black Pattern. TN型液晶面板=>黑色画面； IPS Type=>All White Pattern. IPS型液晶面板=>白色画面； Temperature：25℃；温度：室温25摄氏度；				
Sleep Mode 休眠模式	VCI	-	-	uA
测试条件：VCI=2.8V，IOVCC=1.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. 除IC内部晶振和面板扫描外，其他功能都暂停工作； Temperature：25℃；温度：室温25摄氏度；				

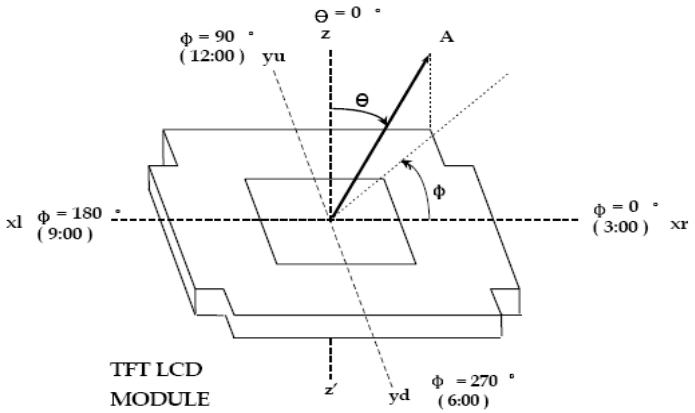
4.Optical Specification

4.1 LCM Optical Characteristics

Item	Conditions		Min.	Typ.	Max.	Unit	Note
Transmittance	Center		(3.8)	(4.3)	-	%	Under C-light (1),(5),(7),(8) θx=θy=0°
Contrast Ratio	Center		(1,000)	(1,200)	-	-	(1),(3),(6),(7),(8) θx=θy=0°
Response Time	Rising + Falling		-	(30)	(35)	ms	(1),(4),(6),(7),(8) θx=θy=0°
CF Color Chromaticity (CIE1931)	Red x		Typ. (-0.03)	(0.662)	Typ. (+0.03)	-	Under C-light (1),(5),(8) θx=θy=0°
	Red y			(0.329)		-	
	Green x			(0.282)		-	
	Green y			(0.599)		-	
	Blue x			(0.136)		-	
	Blue y			(0.115)		-	
	White x			(0.303)		-	
	White y			(0.339)		-	
NTSC	CIE1931		(65)	(70)	-	%	
Viewing Angle (CR≥10)	Horizontal	θ_{x+}	(80)	(85)	-	degree	(1),(2),(6),(7),(8)
		θ_{x-}	(80)	(85)	-		
	Vertical	θ_{y+}	(80)	(85)	-		
		θ_{y-}	(80)	(85)	-		

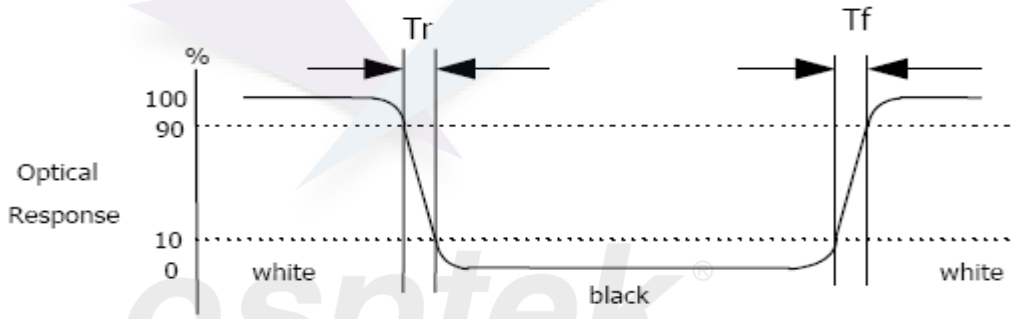
4.2 Measurement system

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

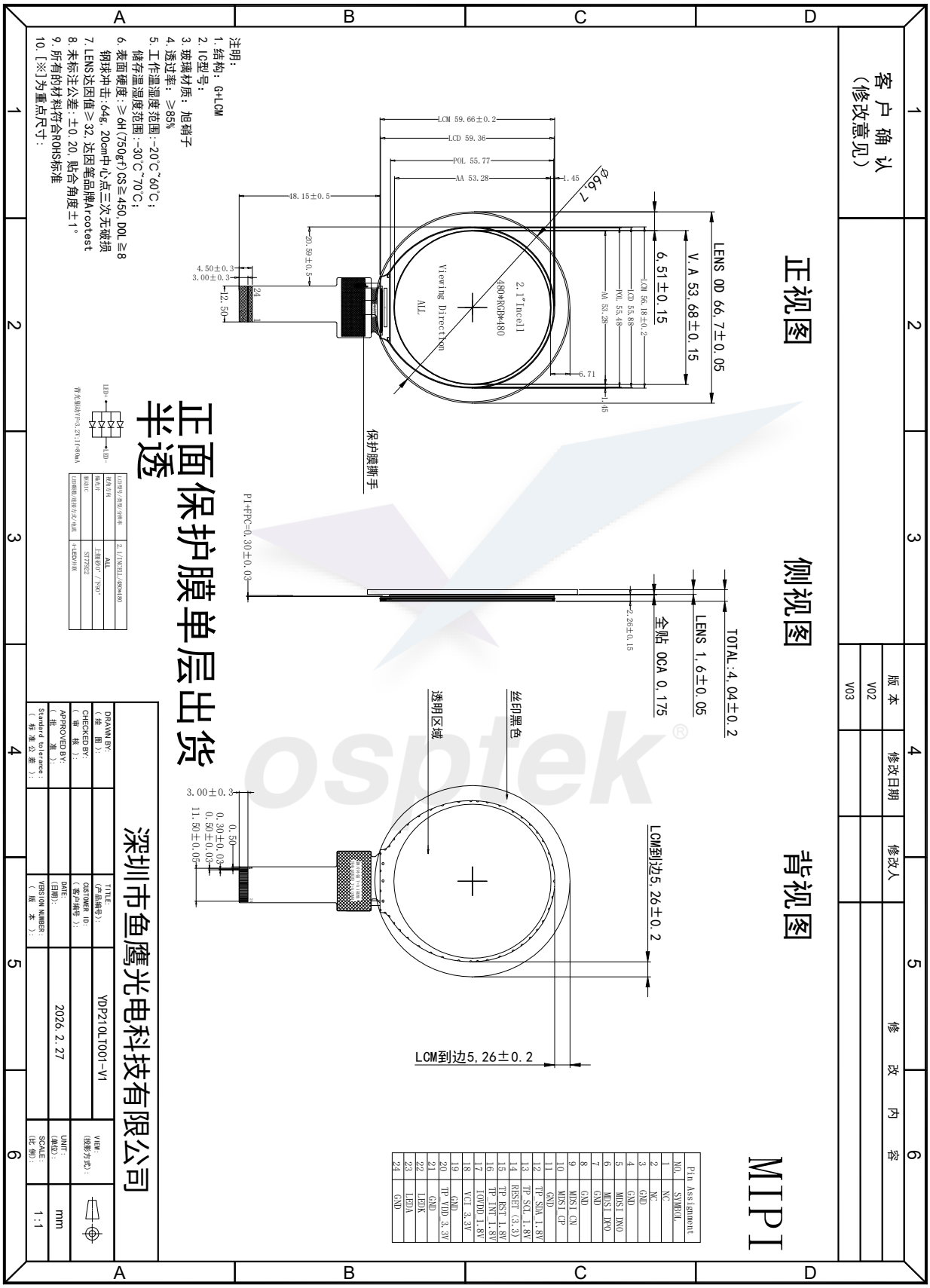
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing



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6. Reliability Test Items

Test Item	Test Condition	Test result determinant gist
High temperature storage	80℃，24H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the LCD; 2.Non-display; 3.Glass crack; 4. The electrical characteristics requirements shall be satisfied.
Low temperature storage	-30℃±3℃，24H；	
High temperature operation	70℃±℃，24H；	
Low temperature operation	-20±℃，24H；	
High temperature /humidity	60℃±3℃,90%±3%RH，24H；	
Thermal Shock	-30℃/0.5h~+80℃/0.5h for a total 24 cycles；	
Vibration Test	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H；(Packing condition)	
ESD test	±2KV, Human Body Mode, 150pF/330Ω； ±8KV, Air Mode, 150pF/330Ω；	

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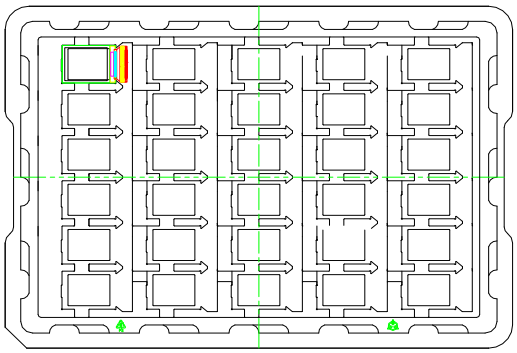
Remark:

1. The test samples should be applied to only one test item.
2. Sample size for each test item is 2pcs.
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

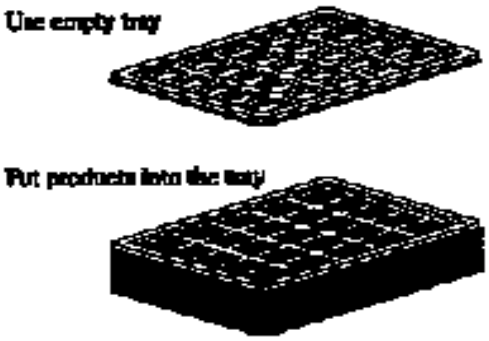
7. Packing and Storage Specification(Reference Only)

7.1 Packing Method

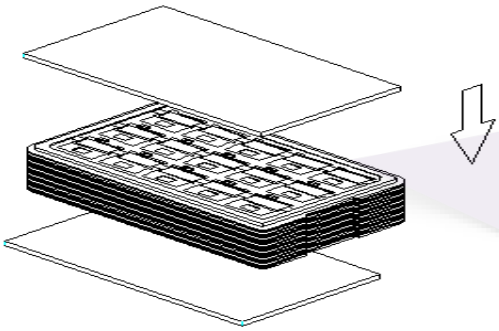
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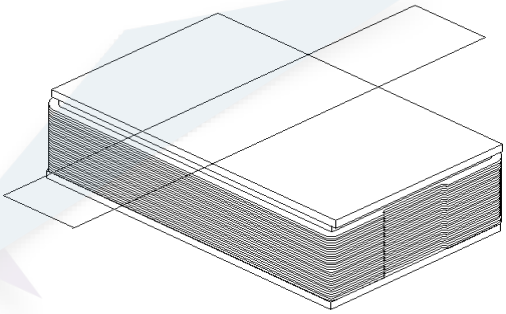
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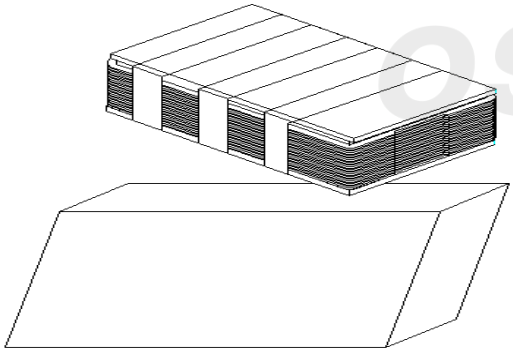
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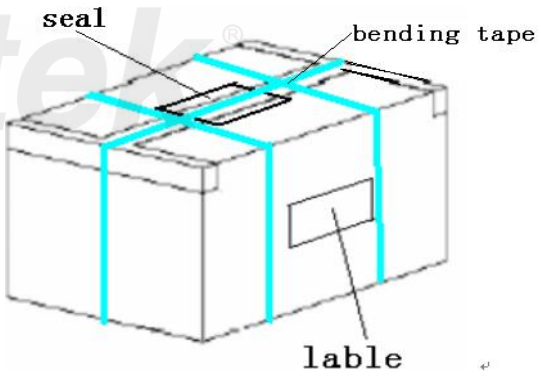
(4)



(5)



(6)



1. Put module into tray cavity. .
2. Tray stacking. .
3. Put 1 foam under the tray stack and 1 foam above. .
4. Fix the cardboard to the tray stack with adhesive tape. .
5. Put the tray stack into carton. .
6. Carton sealing with adhesive tape. .

7.2 Storage Method

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.

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8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.

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